

7.3 Administration

AI tools also help with the administration of DBMSs. Whether it's more trivial tasks such as setting up new databases or performing a backup, or whether you need to set up and configure a DBMS from scratch, ChatGPT and others are there to help you with tips.

In our experience, the quality of such instructions is usually good in simple cases. If, on the other hand, you need real specialist knowledge where a lengthy internet search will only yield a few and possibly contradictory pieces of advice, you should also treat AI answers with a great deal of caution. [Section 7.3.4](#) provides an example of this.

7.3.1 PostgreSQL: User Administration

The setup of new databases always involves the same steps: create database, create users, and grant the relevant privileges to the users. In this respect, AI tools can save you a lot of typing.

Prompt: PostgreSQL: I want to create a new database and set up three users: one with full administrative privileges, one for database use (basically SELECT, INSERT, UPDATE, DELETE), and one for an automated backup system. Can you help me with the SQL commands I need?

Certainly! Below are the SQL commands needed to create a new database and set up three users with the specified privileges in PostgreSQL [more details]

- Connect to PostgreSQL as a superuser (usually postgres).

```
psql -U postgres
```

- Create the database:

```
CREATE DATABASE your_database_name;
```

- Create the users:

```
CREATE USER admin_user    WITH PASSWORD 'admin_pass-  
word';  
CREATE USER regular_user WITH PASSWORD 'regular_-  
password';  
CREATE USER backup_user   WITH PASSWORD 'backup_-  
password';
```

- Grant privileges:

```
-- Admin user  
GRANT ALL PRIVILEGES ON DATABASE your_database_name  
    TO admin_user;  
\c your_database_name  
GRANT ALL PRIVILEGES ON ALL TABLES IN SCHEMA public  
    TO admin_user;  
...  
ALTER DEFAULT PRIVILEGES IN SCHEMA public  
    GRANT ALL PRIVILEGES ON FUNCTIONS TO admin_user;  
  
-- Regular user  
GRANT CONNECT ON DATABASE your_database_name  
    TO regular_user;  
...  
  
-- Backup user  
GRANT CONNECT ON DATABASE your_database_name  
    TO backup_user;  
...
```

Replace `your_database_name`, `admin_user`, `admin_password`, `regular_user`, `regular_password`, `backup_user`, and `backup_password` with your desired names and passwords.

There is little to criticize in the answer, which is somewhat abbreviated here for reasons of space. The difficulty of this question wasn't high. The

primary purpose of an AI tool here is to save time. This works even more efficiently if you enter the desired names for the database and the user accounts.

Prompt: Perfect. Could you please repeat the SQL commands (without the explanation) using the following names:

database: bikerental

admin user: bike_admin

regular user: bike_user

backup user: bike_backup

7.3.2 SQL Server: Automated Backup

The classic administration task is to run a backup. Depending on the DBMS and the database size, there are countless variants. In this example, we want to set up an incremental backup process.

Prompt: SQL Server: I would like to set up an automated backup system for my accounting database: a full backup every Sunday night, incremental backups the rest of the week. I want the system to keep the last two weeks of backups before overwriting them. The backups should be stored in a local directory, from where they are synchronized to an external server (not relevant here).

As ChatGPT's answer would be three and a half pages long in this book, we've only quoted the key points here.

To set up an automated backup system for your SQL Server database with the specified requirements, you can use SQL Server Agent for scheduling the backup jobs. Here's a step-by-step guide:

Create a script for the full backup, saving it as FullBackup.sql:

```
DECLARE @BackupFileName NVARCHAR(500)
DECLARE @BackupPath NVARCHAR(500) = 'C:\SQLBackups\'
DECLARE @DatabaseName NVARCHAR(500) =
    'YourDatabaseName'
-- Generate file name with timestamp
```

```

SET @BackupFileName = @BackupPath + @DatabaseName +
    '_Full_' + CONVERT(NVARCHAR(500), GETDATE(), 112) +
    '.bak'
-- Perform full backup
BACKUP DATABASE @DatabaseName
TO DISK = @BackupFileName
WITH INIT

```

Create a script for the differential backup, saving it as DifferentialBackup.sql:

```

...
-- Perform differential backup
BACKUP DATABASE @DatabaseName
TO DISK = @BackupFileName
WITH DIFFERENTIAL, INIT

```

Create a script for cleaning up old backups, saving it as CleanupBackups.sql:

```

DECLARE @BackupPath NVARCHAR(500) = 'C:\SQLBackups\'
DECLARE @DaysToKeep INT = 14
-- Delete backup files older than @DaysToKeep days
EXECUTE master.dbo.xp_delete_file 0, @BackupPath,
    'bak', @DaysToKeep, 1

```

Create SQL Server Agent jobs: Open SQL Server Management Studio (SSMS), connect to your SQL Server instance, expand the SQL Server Agent node, right-click on Jobs, and select New Job...

The solution sounds reasonable, and you can implement the backup system. You'll want to take a look at the original documentation:

Prompt: Can you provide manual links for BACKUP and RESTORE DATABASE?

Certainly! Here are the links to the official Microsoft documentation for the BACKUP and RESTORE DATABASE commands in SQL Server:

- <https://docs.microsoft.com/en-us/sql/t-sql/statements/backup-transact-sql?view=sql-server-ver15>

- <https://docs.microsoft.com/en-us/sql/t-sql/statements/restore-statements-transact-sql?view=sql-server-ver15>

The links refer to SQL Server 15 (2019). SQL Server 16 (2022) was the current version at the time we tested this example. However, no major syntax changes are to be expected for the backup and restore commands; in this respect, the links are OK.

The most important rule when setting up a backup system is to test the recovery too!

Prompt: Suppose my main server goes down on a Wednesday. I pull the backups from my off-site backup source. How do I recover my database?

To recover your database from backups when the main server goes down, follow these steps [more details]. Here's a summary of all the commands you might need:

```
-- Restore full backup
RESTORE DATABASE YourDatabaseName
FROM DISK =
'C:\SQLBackups\Accounting_Full_20240602.bak'
WITH NORECOVERY;
-- Restore differential backup from June 3, 2024
RESTORE DATABASE Accounting
FROM DISK =
'C:\SQLBackups\Accounting_Diff_20240603.bak'
WITH NORECOVERY;
-- Restore differential backup from June 4, 2024
RESTORE DATABASE Accounting
FROM DISK =
'C:\SQLBackups\Accounting_Diff_20240604.bak'
WITH NORECOVERY;
-- Restore differential backup from June 5, 2024
RESTORE DATABASE Accounting
FROM DISK =
'C:\SQLBackups\Accounting_Diff_20240605.bak'
WITH RECOVERY;
```

7.3.3 Determining the Schema of a Database

For ChatGPT to help you with questions about an existing database, you must first make the database schema known. The easiest way to do this is to have an SQL file with all the required `CREATE TABLE` commands. With ChatGPT, you can upload the file using drag and drop. In VS Code or another editor, you can load the file and refer to this file in your question.

But where do you get such a file from? Almost all DBMSs have commands to generate corresponding schema files in text form. The easiest way to do this is to ask ChatGPT about the program you're using.

Prompt: MySQL: How can I save my database schema in a SQL file?

To save your MySQL database schema to a SQL file, you can use the `mysqldump` command with the `--no-data` option. This will create a SQL file containing only the database structure (tables, indexes, etc.) without any of the data.

```
mysqldump -u [username] -p --no-data [dbname] > schema.sql
```

Here's a breakdown of the command [\[more details\]](#)

Prompt: Is there a similar command for SQL Server?

Yes, in SQL Server, you can use the SQL Server Management Studio (SSMS) or a script to generate the schema-only backup. Here's how you can do it using both methods:

Using SQL Server Management Studio (SSMS)

- Open SSMS and connect to your SQL Server instance.
- Right-click the database you want to script.
- Select **Tasks • Generate Scripts**.
- In the wizard, click **Next** to go past the introduction screen.
- Select the specific database objects you want to script (or select the entire database). Click **Next**.
- On the **Set Scripting Options** page, click **Advanced**.

- [more details]

Using a Script with SQLCMD

Alternatively, you can use the `sqlcmd` utility along with a custom query to generate the schema-only script. Here's an example script:

```
sqlcmd -S [server_name] -U [username] -P [password] `
-d [database_name] `
-Q "EXEC sp_generate_inserts @schema_only = 1" `
-o "[output_file].sql"
```

Using PowerShell and SQL Server Management Objects (SMO)

You can also use a PowerShell script to generate the schema. [more details]

With MySQL, the `mysqldump` command is the easiest way. For SQL Server, ChatGPT provides a choice of three solutions, the first of which relates to *SQL Server Management Studio* (SSMS). However, such instructions have the disadvantage that they can often only be reproduced in an identical way if you use a specific version of this user interface.

Prompt: To which version of SSMS does your answer apply?

The steps provided for generating a schema-only script using SQL Server Management Studio (SSMS) apply to SSMS versions 2012 and later, including the latest versions up to SSMS 19. SSMS has had a consistent interface for these scripting tasks across these versions, so the steps should be applicable.

However, when we ran the preceding prompt in May 2024, version 20.1 was already available, while ChatGPT's training material wasn't that up to date. But with a bit of luck, the instructions will work in the same or a similar way with the latest version.

7.3.4 MariaDB Tuning

A few months ago, we set up a LAMP server (Linux + Apache + MariaDB + PHP). MariaDB has a major impact on the overall performance of the web application running there. Accordingly, we've put a lot of effort into tuning the database server. Would ChatGPT have gotten us there?

Prompt: I have just installed a new LAMP server running Ubuntu 24.04. The server has 64 GB of RAM. It will run as a web server (Apache), mail server (Postfix), and database server (MariaDB 10.11). I need help tuning MariaDB.

MariaDB may use up to 50% of the RAM. There are many databases (about 200) with about 20 tables each. Read performance is critical. (The ratio between SELECT and INSERT/UPDATE/DELETE is about 100:1.)

Please give me five concise recommendations for MariaDB or Linux settings to achieve good performance.

To optimize MariaDB for your LAMP server setup, follow these five recommendations:

1. Adjust the InnoDB Buffer Pool Size [more details]

```
[mysqld]
innodb_buffer_pool_size = 32G
```

2. Optimize Query Cache [more details]

```
[mysqld]
query_cache_type = 1
query_cache_size = 1G
query_cache_limit = 1M
```

3. Increase the Number of InnoDB Buffer Pool Instances [more details]

```
[mysqld]
```



```
innodb_buffer_pool_instances = 16
```

4. Tune InnoDB Log File Size and Flush Method [\[more details\]](#)

```
[mysqld]  
innodb_log_file_size = 2G  
innodb_flush_method = O_DIRECT
```

5. Optimize File System and I/O Scheduler: Use the XFS or ext4 file system with the `noatime` option to reduce disk I/O and set the I/O scheduler to deadline for better performance. [\[more details\]](#)

```
sudo mount -o remount,noatime /  
sudo echo deadline > /sys/block/sda/queue/scheduler
```

We weren't entirely convinced by the proposals. Only the first measure is truly uncontroversial.

We also enabled the query cache, independent of ChatGPT's recommendation. However, in recent years, the prevailing opinion has been that the cache positively impacts overall performance only in exceptional cases. A look at the cache statistics suggests that our server is indeed one of these exceptions. Be that as it may, in current MySQL versions, the query cache is no longer available; in MariaDB, it's disabled by default. ChatGPT's recommendation for the query cache likely results from countless old websites that were used in training the language model. This is quite a common problem; ChatGPT tends to prefer outdated measures because they appear more frequently in the training material than modern guidelines.

Measures 3, 4, and 5 are unlikely to result in any measurable differences in performance. If you really want to use the deadline scheduler, note that it's called `mq-deadline` in current kernel versions (including Ubuntu 24.04, which was mentioned in the prompt).

However, we're missing another measure in the list of suggestions, which would actually be obvious with the setup described in the prompt: increas-

ing the limit for simultaneously open files (`mysql soft nfile 65535` and `mysql hard nfile 65535` in `/etc/security/limits.conf`).

In short, as a tuning aid, ChatGPT can provide food for thought, but the quality of the response is nowhere near as good as when compiling SQL commands.